CIT 140 · Week 7 · Data-Quality Report

Meridian Forge · fictional inspection export v1

Learner name / date	Kristen Ramirez 11/9/2026
Import status	20 records imported into the Prepared worksheet using Power Query.

Amber cells: your responses. Other results calculate from Prepared and Audit.

Rule	Check	Rows flagged	Share of all rows	**Affected Row_IDs**
Q01	Machine ID is required	2	10%	R03, R14
Q02	Work order is unique	2	10%	R05, R06
Q03	Status uses the agreed list	2	10%	R08, R16
Q04	Duration is in range	3	15%	R10, R18, R20
Q05	READY identifies an inspector	3	15%	R03, R07, R12
Distinct rows with any issue		11	55%	Count each row once, even when several Q-rules flag it. Do not add the five counts.

Scope: all 20 supplied records. Q02 flags every row in a repeated Work_Order_ID group; extra duplicate copies are a different measure. Unresolved fields remain flagged, not guessed.

Source: authored CIT140 case W07-S01 (Raw) and W07-S02 (rules). Report a data-quality finding, not a machine safety clearance. Explain your evidence and next action on page 2.

Week 7 · Explanation and refresh evidence

Finding: Row_ID, Q-number, actual value, source and effect

In the Week 7 inspection records, R03 and R14 are missing Machine_ID values, which violates Q01. R08 has an invalid status of RDEAY, which violates Q03. These issues could make it difficult to identify equipment correctly or determine the status of an inspection. The findings are supported by W07-S01 and W07-S02.

Next action: what to verify, who owns it, and how to retest

The data owner should verify the missing machine IDs against the original inspection records and confirm the correct status for R08. After correcting the source data, I would refresh Power Query and check the Audit sheet again to make sure the flags update.

Limit: why you cannot guess a correction or prove accuracy

I cannot assume what the missing or incorrect values should be because the data does not provide enough evidence. The validation rules can identify problems, but they cannot prove that the remaining information is accurate.

Professional connection: one framework idea and how you used it

I used the data-quality principle of validity by checking whether each record followed the required rules. I also used completeness to identify missing information and uniqueness to find repeated work orders.

Refresh test: before, changed value, after, and restored result

I tested the refresh process using R08. The original status was RDEAY, which was flagged by Q03. I temporarily changed it to READY and refreshed the data to verify that the flag disappeared. I then restored RDEAY and refreshed again. The Q03 flag returned, confirming that the audit results update when the source data changes.

Export only Report: two landscape pages, page break before row 26. Review both pages.